College of Computer and Information Sciences

CS207

Assignment2

Computer Sciences Department

Name	Ebtisam Bassam	Academic No.	446008139	Section	4c2

Assignment #2

FOR INSTRUCTOR'S USE ONLY

Q.No.	Max. Marks	Marks Obtained
1	8	
Total Score	8	

General Instructions:

1. Each student has to solve the assignment **individually.**
2. The assignment, after writing the solutions, should be saved as a PDF file and the Design# as a compassed file.
3. The solution should be submitted via the blackboard.
4. Submissions are not allowed after the deadline.
5. Copying solutions from other individuals will result in a mark of ZERO for the assignment.
6. **Assigning a mark to the assignment will only happen after conferring with the student individually.**
7. Due Date: end of Week 11.

The objective of this assignment is to design a 4-bit comparator using Verilog. A comparator is a digital circuit that compares two binary values and determines their relationship (greater than, less than, or equal). This project will enhance your understanding of combinational logic circuits and the Verilog hardware description language.

Requirements

1. Verilog Module

You are required to create a module named Comparator that includes:

- Two 4-bit inputs (A and B)
- Three output signals that indicate the comparison results:
    - A_greater (1 if A > B)

- A_equal (1 if A == B)
- A_less (1 if A < B)

2. Functionality

The comparator should:

- Compare the two 4-bit inputs and set the appropriate output signals.

3. Schematic Diagram

Draw a simple schematic diagram that illustrates:

- The two 4-bit inputs
- The output signals
- The internal logic used for comparison

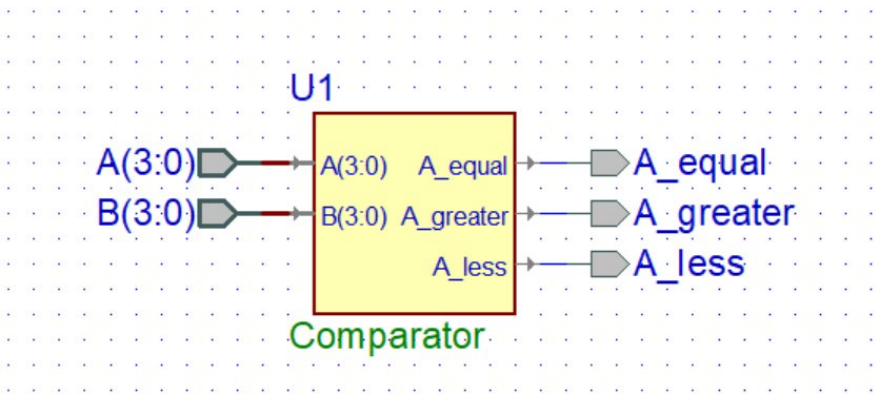
4. Results Analysis

Analyze the results obtained from the comparator, focusing on:

- Correctness of the comparison results.
- Discussion on the applications of comparators in digital systems.

Solution

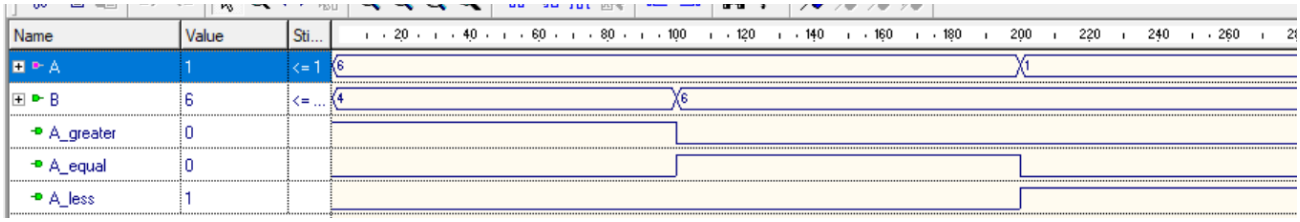
Diagram (Screenshot) (2 Marks):

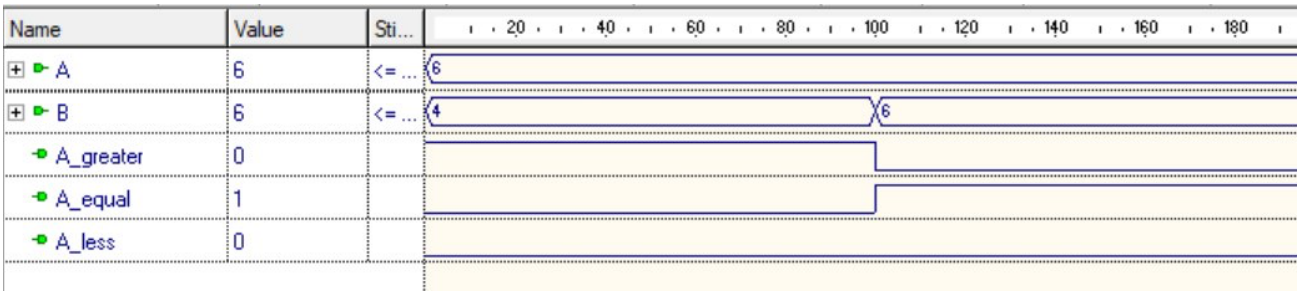


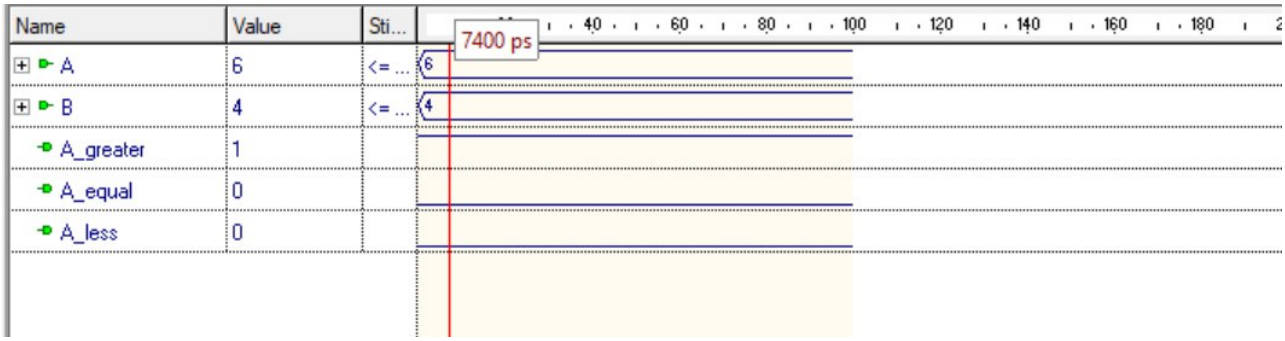
Code (Screenshot) (2 Marks):

```
module Comparator (A, B, A_greater, A_equal, A_less);

input [3:0] A;
input [3:0] B;

output reg A_greater;
output reg A_equal;
output reg A_less;

always @(A or B) begin
    if (A > B) begin
        A_greater = 1;
        A_equal   = 0;
        A_less    = 0;
    end

    else if (A == B) begin
        A_greater = 0;
        A_equal   = 1;
        A_less    = 0;
    end

    else if (A < B) begin
        A_greater = 0;
        A_equal   = 0;
        A_less    = 1;
    end
end

endmodule
```

Input and output ((2 Marks):







Brief explanation (Use your own words to write the explanation) (2 Marks):

This design implements a 4-bit comparator using Verilog. The module takes two 4-bit inputs (A and B) and compares them using an always block based on combinational logic.

The comparison is done using if-else conditions. If A is greater than B, the signal A_greater is set to 1 while the other outputs are set to 0. If A is equal to B, A_equal becomes 1. Otherwise, if A is less than B, A_less is set to 1.

The always block is sensitive to changes in inputs A and B, so the output updates immediately whenever the inputs change. The simulation results confirm that the comparator works correctly for all cases (greater than, equal, and less than).

Comparators are widely used in digital systems such as processors, sorting circuits, and control units to make decisions based on data comparison.